Detailed Results on the New Instances

Supplement to W. Wang, "Enhanced Exact Solution of the Electric Vehicle Routing and Overnight Charging Scheduling Problem"

Tables 1–5 report the results of our branch-price-and-cut algorithm on each of the 280 new instances with 60, 70, 80, 90, and 100 customers. They are the runs of Section 6.4 of the paper: Java 17 and Gurobi 12.0.0 on one core of an AMD EPYC 9655 processor of the Fir cluster, with 32 GB of memory and a time limit of 7,200 seconds that includes the initial heuristic. The logs of these runs are in the folder `results/new/ours` of the repository, and this document is generated from them.

The columns are the following.

- B: number of chargers.
- Init. UB: value of the solution of the initial heuristic; a dash indicates that the heuristic found no feasible solution within its time limit.
- UB: value of the best solution found.
- LB: global lower bound at the end of the run; a dash indicates that the root node was not completed.
- Gap (%): $100(\mathrm{UB}-\mathrm{LB})/\mathrm{UB}$.
- Time (s), Total: computing time, including the initial heuristic; TL indicates that the time limit was reached.
- Time (s), Best: time at which the best solution was found.
- Nodes: number of processed branch-and-bound nodes.
- Closed by, Enum.: number of nodes solved by the charging-flexible models over the paths enumerated with the whole gap (Section 5.2).
- Closed by, Cert.: number of nodes closed by the certificate of the enumeration ladder (Proposition 7).
- Veh.: number of vehicles in the best solution.

Table 1 Results on the 56 instances with 60 customers (53 solved to optimality).

		Bounds				Time (s)			Closed by		
Instance	B	Init. UB	UB	LB	Gap (%)	Total	Best	Nodes	Enum.	Cert.	Veh.
C101-60	2	510.2	499.2	499.20	0.00	26.5	11.0	1	0	0	7
C102-60	2	480.8	468.0	468.00	0.00	17.7	17.7	1	0	0	6
C103-60	2	468.0	468.0	468.00	0.00	33.9	11.0	1	0	0	6
C104-60	2	467.7	464.6	464.60	0.00	98.6	98.6	1	0	0	6
C105-60	2	498.9	469.0	469.00	0.00	17.5	17.5	1	0	0	6
C106-60	2	489.0	470.9	470.90	0.00	15.2	15.2	1	0	0	6
C107-60	2	469.0	469.0	469.00	0.00	11.4	8.7	1	0	0	6
C108-60	2	469.0	469.0	469.00	0.00	10.3	6.8	1	0	0	6
C109-60	2	469.0	469.0	469.00	0.00	15.7	8.1	1	0	0	6
C201-60	1	474.7	446.4	446.40	0.00	59.5	58.3	1	0	1	4
C202-60	1	446.4	446.4	446.40	0.00	104.7	13.6	1	0	0	4
C203-60	1	493.9	446.4	446.40	0.00	458.7	458.6	1	0	0	4
C204-60	1	495.4	444.6	444.60	0.00	2500.3	2500.3	1	0	0	4
C205-60	1	482.9	446.4	446.40	0.00	63.8	63.8	1	0	0	4
C206-60	1	740.4	446.4	446.40	0.00	54.1	54.1	1	0	0	4
C207-60	1	480.9	444.8	444.80	0.00	178.5	178.5	1	0	0	4
C208-60	1	490.8	446.2	446.20	0.00	111.7	83.7	1	0	0	4
R101-60	5	–	1159.7	1159.70	0.00	15.5	10.3	1	0	1	14
R102-60	5	–	1002.5	1002.50	0.00	8.2	7.0	1	0	1	12
R103-60	5	–	820.7	820.70	0.00	23.6	23.6	1	0	0	10
R104-60	5	725.9	677.3	677.30	0.00	127.5	122.4	7	3	0	7
R105-60	5	1023.5	981.0	981.00	0.00	34.8	34.8	1	0	1	11
R106-60	5	890.7	862.2	862.20	0.00	31.5	21.9	1	0	1	9
R107-60	5	778.2	767.4	767.40	0.00	41.2	41.2	1	0	0	8
R108-60	5	696.4	654.1	654.10	0.00	39.5	39.5	1	0	0	7
R109-60	5	851.1	837.3	837.30	0.00	24.9	24.9	1	0	0	9
R110-60	5	795.4	774.3	774.30	0.00	34.8	25.7	1	0	1	8
R111-60	5	770.9	753.9	753.90	0.00	36.2	26.0	1	0	1	8
R112-60	5	718.4	686.0	686.00	0.00	66.4	66.4	1	0	1	7
R201-60	1	1020.1	964.8	964.80	0.00	141.3	135.1	1	0	1	10
R202-60	1	894.0	831.5	831.50	0.00	396.5	249.9	1	1	0	8
R203-60	1	747.3	727.5	704.70	3.13	TL	433.7	2	0	0	7
R204-60	1	667.8	610.0	610.00	0.00	509.0	509.0	1	0	0	6
R205-60	1	847.1	808.5	808.50	0.00	698.0	110.3	3	2	0	8
R206-60	1	796.1	729.5	729.50	0.00	678.7	612.5	1	1	0	7
R207-60	1	720.2	675.2	663.60	1.72	TL	692.4	2	0	0	6
R208-60	1	666.9	599.6	599.60	0.00	138.3	138.3	1	0	0	5
R209-60	1	748.7	707.5	707.50	0.00	1010.6	1010.6	3	2	0	7
R210-60	1	850.4	727.8	727.80	0.00	1367.5	1341.2	1	0	1	7
R211-60	1	682.1	623.1	611.90	1.80	TL	915.2	2	0	0	6
RC101-60	5	–	1239.4	1239.40	0.00	57.0	57.0	1	0	1	12
RC102-60	5	1050.7	1050.7	1050.70	0.00	16.3	5.5	1	0	0	10
RC103-60	5	1010.9	949.9	949.90	0.00	94.3	94.3	1	0	0	9
RC104-60	5	915.2	849.0	849.00	0.00	2231.0	94.5	545	199	0	8
RC105-60	5	1113.2	1072.7	1072.70	0.00	47.8	45.8	1	0	1	10
RC106-60	5	–	1015.0	1015.00	0.00	168.8	43.8	3	2	0	10
RC107-60	5	–	897.6	897.60	0.00	834.5	209.0	91	42	0	9
RC108-60	5	871.7	825.4	825.40	0.00	350.5	148.5	19	8	0	8
RC201-60	2	1128.8	1045.3	1045.30	0.00	29.8	29.8	1	0	0	9
RC202-60	2	1025.7	916.9	916.90	0.00	25.9	25.9	1	0	0	8
RC203-60	2	922.2	872.4	872.40	0.00	43.2	43.2	1	0	0	8
RC204-60	2	674.6	670.6	670.60	0.00	80.3	80.3	1	0	0	6
RC205-60	2	1003.5	999.4	999.40	0.00	47.6	39.6	1	0	1	9
RC206-60	2	945.3	936.7	936.70	0.00	61.2	61.2	1	0	0	8
RC207-60	2	798.4	794.8	794.80	0.00	32.5	17.6	1	0	0	7
RC208-60	2	667.3	664.8	664.80	0.00	446.5	43.5	1	0	0	6

Table 2 Results on the 56 instances with 70 customers (51 solved to optimality).

		Bounds				Time (s)			Closed by		
Instance	B	Init. UB	UB	LB	Gap (%)	Total	Best	Nodes	Enum.	Cert.	Veh.
C101-70	2	682.0	675.1	675.10	0.00	54.7	53.5	3	1	0	9
C102-70	2	699.8	642.3	642.30	0.00	41.9	41.9	1	0	1	8
C103-70	2	697.9	615.8	615.80	0.00	65.1	59.0	1	0	1	7
C104-70	2	683.5	683.5	–	–	TL	18.1	1	0	0	8
C105-70	2	777.2	640.2	640.20	0.00	26.0	26.0	1	0	0	8
C106-70	2	676.8	638.2	638.20	0.00	26.6	17.6	1	0	0	8
C107-70	2	636.3	620.4	620.40	0.00	22.4	19.6	1	0	1	7
C108-70	2	615.6	614.3	614.30	0.00	53.0	51.3	1	0	1	7
C109-70	2	612.3	612.3	612.30	0.00	20.0	11.6	1	0	0	7
C201-70	1	621.4	616.7	616.70	0.00	5809.9	5809.9	3	1	0	6
C202-70	1	623.8	609.7	609.70	0.00	1935.4	1163.6	3	1	0	5
C203-70	1	599.5	599.0	597.40	0.27	TL	6365.6	12	3	0	5
C204-70	1	589.2	581.2	581.20	0.00	3309.5	3309.5	1	0	0	5
C205-70	1	614.4	607.8	601.10	1.10	TL	6199.5	2	0	0	5
C206-70	1	612.3	596.0	596.00	0.00	1584.9	1584.9	1	0	0	5
C207-70	1	611.0	594.4	594.40	0.00	3958.9	2764.8	3	0	0	5
C208-70	1	620.4	595.8	595.80	0.00	3727.8	3727.8	1	0	0	5
R101-70	7	–	1398.4	1398.40	0.00	15.0	13.9	1	0	1	17
R102-70	7	–	1264.2	1264.20	0.00	11.2	10.3	1	0	1	15
R103-70	7	–	1013.1	1013.10	0.00	20.4	20.4	1	0	1	12
R104-70	7	849.1	789.4	789.40	0.00	61.7	59.5	1	0	1	8
R105-70	7	1201.5	1153.0	1153.00	0.00	37.0	25.3	5	3	0	13
R106-70	7	–	1061.9	1061.90	0.00	32.3	32.3	1	0	1	12
R107-70	7	937.8	916.3	916.30	0.00	57.9	56.1	1	0	1	9
R108-70	7	817.4	778.5	778.50	0.00	170.8	94.6	5	0	0	8
R109-70	7	1001.2	996.0	996.00	0.00	47.4	45.4	1	0	1	11
R110-70	7	1001.5	894.6	894.60	0.00	32.0	23.4	1	0	0	9
R111-70	7	1174.3	891.7	891.70	0.00	43.8	43.8	1	0	0	9
R112-70	7	817.1	806.3	806.30	0.00	109.0	102.0	1	0	1	8
R201-70	2	1067.7	1016.8	1016.80	0.00	77.9	74.8	3	0	0	9
R202-70	2	1014.4	921.3	921.30	0.00	86.8	86.8	1	0	0	9
R203-70	2	873.0	808.8	808.80	0.00	138.9	138.9	1	0	1	7
R204-70	2	749.5	709.9	709.90	0.00	112.9	112.9	1	0	0	6
R205-70	2	961.7	900.7	900.70	0.00	93.6	93.6	1	0	0	8
R206-70	2	864.7	839.1	839.10	0.00	136.4	136.4	1	0	0	7
R207-70	2	865.9	788.1	776.10	1.52	TL	195.0	2	0	0	7
R208-70	2	756.4	708.0	708.00	0.00	188.5	188.5	1	0	0	6
R209-70	2	864.2	805.5	805.50	0.00	279.8	168.4	23	1	0	7
R210-70	2	988.5	842.7	842.70	0.00	179.0	179.0	1	0	1	7
R211-70	2	768.4	725.3	725.30	0.00	1032.0	1032.0	9	0	0	6
RC101-70	5	–	1492.9	1492.90	0.00	48.9	33.3	1	0	1	16
RC102-70	5	–	1198.1	1198.10	0.00	45.9	44.5	1	0	1	12
RC103-70	5	1091.1	1070.8	1070.80	0.00	982.6	348.5	59	25	0	10
RC104-70	5	906.7	872.9	872.90	0.00	3703.6	3703.1	27	11	0	8
RC105-70	5	–	1242.0	1242.00	0.00	83.3	69.3	1	0	1	13
RC106-70	5	1174.5	1096.7	1096.70	0.00	768.8	103.8	57	29	0	10
RC107-70	5	1007.3	959.5	949.80	1.01	TL	6770.8	513	16	0	9
RC108-70	5	898.4	866.3	866.30	0.00	4362.9	3798.5	57	24	0	8
RC201-70	2	1154.5	1142.0	1142.00	0.00	1445.2	1084.0	131	38	0	11
RC202-70	2	1006.9	990.4	990.40	0.00	156.6	156.6	17	1	0	9
RC203-70	2	924.8	897.1	897.10	0.00	237.1	237.1	1	0	0	8
RC204-70	2	756.3	698.6	698.60	0.00	373.5	373.5	1	0	0	6
RC205-70	2	1104.4	1028.6	1028.60	0.00	91.9	49.8	1	0	0	9
RC206-70	2	1005.1	942.9	942.90	0.00	185.9	185.9	1	0	0	8
RC207-70	2	869.4	807.1	807.10	0.00	140.9	140.9	1	0	0	7
RC208-70	2	699.2	691.6	691.60	0.00	1045.0	1045.0	1	0	0	6

Table 3 Results on the 56 instances with 80 customers (40 solved to optimality).

		Bounds				Time (s)			Closed by		
Instance	B	Init. UB	UB	LB	Gap (%)	Total	Best	Nodes	Enum.	Cert.	Veh.
C101-80	2	932.5	813.6	813.60	0.00	229.2	226.0	13	7	0	12
C102-80	2	847.7	748.2	748.20	0.00	336.2	336.2	1	0	0	10
C103-80	2	699.3	681.1	681.10	0.00	191.6	123.2	1	0	1	8
C104-80	2	697.6	669.8	669.80	0.00	1114.1	1015.9	1	1	0	8
C105-80	2	854.8	734.1	734.10	0.00	166.8	92.6	3	1	0	9
C106-80	2	752.6	710.5	710.50	0.00	203.3	169.5	3	0	0	9
C107-80	2	707.2	687.2	687.20	0.00	36.0	34.8	1	0	1	8
C108-80	2	687.2	687.2	687.20	0.00	107.3	39.3	7	1	0	8
C109-80	2	702.5	676.6	676.60	0.00	29.6	29.6	1	0	0	8
C201-80	1	730.1	710.2	710.20	0.00	164.4	156.0	1	0	0	6
C202-80	1	797.1	708.0	708.00	0.00	852.6	767.1	1	1	0	6
C203-80	1	748.5	702.7	702.40	0.04	TL	1919.7	4	0	0	6
C204-80	1	647.1	647.1	–	–	TL	60.1	1	0	0	6
C205-80	1	650.6	643.8	643.80	0.00	1178.6	1178.6	1	0	0	6
C206-80	1	690.1	643.8	643.50	0.05	TL	4723.0	2	0	0	6
C207-80	1	690.6	640.0	640.00	0.00	5489.8	5433.6	3	0	0	5
C208-80	1	646.5	636.9	632.20	0.74	TL	2615.0	2	0	0	5
R101-80	7	–	1458.7	1458.70	0.00	19.2	16.4	1	0	1	18
R102-80	7	–	1331.1	1331.10	0.00	21.4	19.8	1	0	1	15
R103-80	7	–	1080.6	1080.60	0.00	29.1	24.9	1	0	1	13
R104-80	7	902.4	860.8	860.80	0.00	194.1	181.5	1	0	1	9
R105-80	7	1225.2	1219.3	1219.30	0.00	67.6	64.3	1	0	1	13
R106-80	7	–	1137.7	1137.70	0.00	48.6	48.6	1	0	1	12
R107-80	7	973.0	963.5	963.50	0.00	66.8	57.1	1	0	0	10
R108-80	7	916.7	836.8	836.80	0.00	1329.3	907.2	149	55	0	9
R109-80	7	1066.6	1044.4	1044.40	0.00	63.4	59.6	1	0	1	12
R110-80	7	–	942.1	942.10	0.00	51.4	50.2	1	0	1	10
R111-80	7	962.4	940.2	940.20	0.00	65.2	65.2	1	0	0	10
R112-80	7	894.0	856.8	856.80	0.00	1622.0	224.5	397	169	0	9
R201-80	2	1126.0	1073.8	1073.80	0.00	83.5	83.5	1	0	1	10
R202-80	2	1087.7	991.2	987.90	0.33	TL	491.3	2	0	0	9
R203-80	2	922.5	856.5	848.10	0.98	TL	291.8	2	0	0	8
R204-80	2	773.8	773.1	758.90	1.84	TL	1138.7	2	0	0	7
R205-80	2	978.7	957.6	944.10	1.41	TL	246.9	240	0	0	9
R206-80	2	915.1	877.9	866.00	1.36	TL	215.7	2	0	0	8
R207-80	2	856.4	829.0	808.80	2.44	TL	699.5	2	0	0	7
R208-80	2	808.2	773.8	744.60	3.77	TL	2920.2	2	0	0	7
R209-80	2	887.0	859.3	843.70	1.82	TL	399.5	118	0	0	8
R210-80	2	973.5	898.4	879.50	2.10	TL	363.3	2	0	0	8
R211-80	2	805.3	777.0	755.30	2.79	TL	3755.3	2	0	0	7
RC101-80	7	–	1584.8	1584.80	0.00	78.8	78.8	1	0	1	15
RC102-80	7	–	1406.3	1406.30	0.00	68.7	68.7	1	0	1	13
RC103-80	7	–	1273.2	1273.20	0.00	93.6	73.3	3	1	0	12
RC104-80	7	1216.8	1097.6	1097.60	0.00	6059.9	3750.0	313	141	0	10
RC105-80	7	–	1494.1	1494.10	0.00	136.6	136.3	9	4	0	15
RC106-80	7	1310.9	1277.0	1267.40	0.75	TL	1271.4	1174	178	0	12
RC107-80	7	1256.2	1112.8	1112.80	0.00	115.9	115.9	1	0	0	10
RC108-80	7	1128.8	1079.2	1041.90	3.46	TL	2283.4	17	0	0	10
RC201-80	2	1466.0	1357.3	1357.30	0.00	1809.7	546.9	97	39	0	12
RC202-80	2	1206.9	1158.6	1158.60	0.00	111.7	78.1	1	0	0	10
RC203-80	2	1152.8	1087.6	1087.60	0.00	1095.5	1046.4	59	8	0	10
RC204-80	2	920.5	847.0	847.00	0.00	163.0	162.9	1	0	0	7
RC205-80	2	1275.3	1197.2	1197.20	0.00	132.7	125.8	1	0	1	10
RC206-80	2	1211.0	1120.2	1120.20	0.00	208.6	208.5	1	0	0	9
RC207-80	2	1112.1	991.1	991.10	0.00	106.1	65.2	1	0	0	8
RC208-80	2	953.5	849.5	849.50	0.00	499.2	499.2	1	0	0	7

Table 4 Results on the 56 instances with 90 customers (32 solved to optimality).

		Bounds				Time (s)			Closed by		
Instance	B	Init. UB	UB	LB	Gap (%)	Total	Best	Nodes	Enum.	Cert.	Veh.
C101-90	2	–	1114.2	1114.20	0.00	240.1	67.9	15	4	0	16
C102-90	2	–	1016.0	1016.00	0.00	2466.1	262.7	27	0	0	14
C103-90	2	848.2	770.6	770.60	0.00	824.4	824.4	1	0	0	9
C104-90	2	762.4	752.8	749.40	0.45	TL	7181.1	1	0	0	9
C105-90	2	1123.7	954.2	940.40	1.45	TL	4318.5	684	7	0	14
C106-90	2	918.5	872.4	872.40	0.00	973.9	214.7	65	28	0	11
C107-90	2	871.1	844.5	844.50	0.00	171.9	77.3	5	2	0	11
C108-90	2	814.5	803.4	784.70	2.33	TL	403.3	2	0	0	10
C109-90	2	891.5	753.5	753.50	0.00	54.3	54.3	1	0	0	9
C201-90	1	779.9	775.2	764.10	1.43	TL	1158.0	2	0	0	7
C202-90	1	872.5	764.2	764.20	0.00	4437.2	4409.9	5	2	0	6
C203-90	1	813.9	751.4	749.10	0.31	TL	7096.4	19	1	0	6
C204-90	1	776.4	695.4	668.50	3.87	TL	3571.5	2	0	0	6
C205-90	1	720.5	689.0	689.00	0.00	754.3	754.2	1	0	0	6
C206-90	1	791.0	688.2	688.20	0.00	1024.4	1024.4	1	0	0	6
C207-90	1	729.9	691.3	691.30	0.00	1841.8	1841.8	1	0	0	6
C208-90	1	691.7	688.0	682.60	0.78	TL	3046.7	2	0	0	6
R101-90	10	–	1579.8	1579.80	0.00	16.0	15.7	1	0	1	19
R102-90	10	2048.6	1421.4	1421.40	0.00	20.9	20.9	1	0	0	17
R103-90	10	–	1191.8	1191.80	0.00	39.1	39.1	1	0	1	15
R104-90	10	1488.7	948.6	948.60	0.00	2946.1	2068.3	343	110	0	10
R105-90	10	–	1317.6	1317.60	0.00	31.2	31.2	1	0	1	14
R106-90	10	1255.6	1216.6	1216.60	0.00	67.4	65.6	1	0	1	13
R107-90	10	1068.3	1035.7	1035.70	0.00	106.0	106.0	1	0	1	11
R108-90	10	950.0	914.9	903.00	1.30	TL	232.2	170	0	0	10
R109-90	10	1118.7	1101.9	1101.90	0.00	80.4	79.7	1	0	1	12
R110-90	10	1743.1	1031.8	1031.80	0.00	260.1	126.6	23	7	0	10
R111-90	10	1026.6	1016.4	1016.40	0.00	91.1	77.0	1	0	0	11
R112-90	10	953.1	924.0	919.60	0.48	TL	5634.8	313	0	0	9
R201-90	2	1237.9	1198.2	1169.50	2.40	TL	394.9	170	0	0	12
R202-90	2	1114.1	1069.9	1069.90	0.00	4911.5	2749.3	59	0	0	10
R203-90	2	997.3	900.7	898.60	0.23	TL	970.1	2	0	0	8
R204-90	2	888.0	888.0	–	–	TL	60.1	1	0	0	9
R205-90	2	1061.6	993.5	993.50	0.00	5858.5	415.5	327	65	0	9
R206-90	2	1005.4	927.6	917.30	1.11	TL	672.3	2	0	0	8
R207-90	2	917.1	890.7	847.60	4.84	TL	5464.5	11	0	0	8
R208-90	2	857.9	828.7	777.20	6.21	TL	5269.6	2	0	0	8
R209-90	2	973.5	903.6	892.80	1.20	TL	457.2	2	0	0	8
R210-90	2	975.3	915.4	912.90	0.27	TL	343.8	2	0	0	8
R211-90	2	887.8	861.8	788.80	8.47	TL	5044.7	2	0	0	9
RC101-90	8	–	1677.8	1677.80	0.00	42.9	42.8	1	0	0	18
RC102-90	8	–	1514.5	1514.50	0.00	81.9	72.7	1	0	1	16
RC103-90	8	1343.8	1317.2	1317.20	0.00	492.1	433.2	81	18	0	12
RC104-90	8	1223.4	1171.5	1142.30	2.49	TL	3982.4	6	0	0	11
RC105-90	8	–	1582.1	1582.10	0.00	46.0	41.3	1	0	1	16
RC106-90	8	1472.2	1398.5	1392.50	0.43	TL	428.2	1723	266	0	14
RC107-90	8	1307.5	1241.0	1241.00	0.00	852.6	843.2	129	11	0	12
RC108-90	8	1185.9	1143.8	1123.50	1.77	TL	1094.3	48	0	0	11
RC201-90	2	1708.9	1521.7	1514.00	0.51	TL	2300.3	334	41	0	13
RC202-90	2	1299.1	1267.7	1267.70	0.00	502.9	473.4	15	5	0	11
RC203-90	2	1183.8	1116.3	1116.30	0.00	814.7	797.1	7	1	0	10
RC204-90	2	993.3	888.0	888.00	0.00	1278.1	1278.1	1	0	0	7
RC205-90	2	1391.8	1311.8	1311.80	0.00	838.2	819.9	41	2	0	11
RC206-90	2	1327.0	1255.9	1230.00	2.06	TL	5250.1	145	0	0	11
RC207-90	2	1180.4	1131.6	1113.70	1.58	TL	700.0	53	0	0	10
RC208-90	2	918.8	877.2	877.20	0.00	1551.2	1551.2	1	0	0	7

Table 5 Results on the 56 instances with 100 customers (30 solved to optimality).

		Bounds				Time (s)			Closed by		
Instance	B	Init. UB	UB	LB	Gap (%)	Total	Best	Nodes	Enum.	Cert.	Veh.
C101-100	3	1007.7	995.8	995.80	0.00	113.0	90.6	3	2	0	14
C102-100	3	976.2	901.8	901.80	0.00	238.0	141.5	1	1	0	11
C103-100	3	856.3	851.8	851.80	0.00	140.5	140.4	1	0	0	10
C104-100	3	876.0	846.6	846.60	0.00	1557.7	1421.2	1	1	0	10
C105-100	3	980.9	920.8	920.80	0.00	79.5	79.5	1	0	0	11
C106-100	3	938.4	888.9	888.90	0.00	155.6	88.9	3	0	0	11
C107-100	3	889.0	858.9	858.90	0.00	36.7	36.7	1	0	0	10
C108-100	3	853.7	853.7	853.70	0.00	37.9	26.3	1	0	0	10
C109-100	3	850.9	850.9	850.90	0.00	24.8	12.6	1	0	0	10
C201-100	1	868.7	833.6	822.70	1.31	TL	525.1	103	1	0	8
C202-100	1	883.5	836.2	815.40	2.49	TL	3050.0	2	0	0	8
C203-100	1	841.5	800.3	795.60	0.59	TL	5705.5	2	0	0	7
C204-100	1	866.3	866.3	–	–	TL	60.4	1	0	0	9
C205-100	1	822.2	776.3	776.30	0.00	1577.1	1577.1	1	0	0	6
C206-100	1	810.7	776.3	776.30	0.00	1938.0	1938.0	1	0	0	6
C207-100	1	830.8	774.7	774.70	0.00	2059.9	2059.8	1	0	0	6
C208-100	1	789.4	779.0	771.90	0.91	TL	3327.0	2	0	0	7
R101-100	12	2099.1	1649.8	1649.80	0.00	32.4	32.3	1	0	1	20
R102-100	12	–	1473.3	1473.30	0.00	30.4	30.4	1	0	0	18
R103-100	12	1802.3	1225.3	1225.30	0.00	54.4	39.0	3	1	0	15
R104-100	12	1025.8	985.2	981.50	0.38	TL	1932.9	710	5	0	11
R105-100	12	1959.2	1377.1	1377.10	0.00	263.8	78.7	55	18	0	15
R106-100	12	1799.7	1255.6	1255.60	0.00	1074.8	362.2	493	154	0	13
R107-100	12	1119.5	1074.3	1074.30	0.00	1693.1	1436.9	91	32	0	11
R108-100	12	1408.2	936.5	932.90	0.38	TL	609.8	2	0	0	10
R109-100	12	1196.2	1148.9	1148.90	0.00	133.7	95.6	1	0	1	13
R110-100	12	1120.5	1073.7	1073.70	0.00	234.2	99.6	19	6	0	12
R111-100	12	1668.2	1051.5	1051.50	0.00	212.6	212.6	3	0	0	12
R112-100	12	1152.2	986.4	948.90	3.80	TL	625.3	2	0	0	11
R201-100	2	1256.5	1229.8	1218.40	0.93	TL	554.9	82	0	0	12
R202-100	2	1190.6	1166.9	1095.80	6.09	TL	5270.3	2	0	0	11
R203-100	2	1082.9	1082.9	–	–	TL	61.0	1	0	0	12
R204-100	2	910.7	910.7	–	–	TL	60.2	1	0	0	10
R205-100	2	1147.5	1055.9	1018.90	3.50	TL	894.3	2	0	0	9
R206-100	2	1042.8	1042.8	–	–	TL	60.5	1	0	0	11
R207-100	2	972.5	972.5	–	–	TL	61.2	1	0	0	10
R208-100	2	887.7	887.7	–	–	TL	60.1	1	0	0	9
R209-100	2	1027.3	968.7	927.40	4.26	TL	4117.0	2	0	0	9
R210-100	2	1053.4	1053.4	–	–	TL	60.4	1	0	0	11
R211-100	2	919.8	919.8	–	–	TL	60.2	1	0	0	9
RC101-100	9	–	1732.1	1732.10	0.00	80.7	78.9	5	3	0	17
RC102-100	9	–	1558.3	1558.30	0.00	299.5	261.6	49	11	0	15
RC103-100	9	1458.3	1372.1	1368.80	0.24	TL	2788.2	1457	323	0	13
RC104-100	9	1309.5	1245.4	1191.80	4.30	TL	4194.3	2	0	0	12
RC105-100	9	1685.1	1643.3	1643.30	0.00	110.6	109.7	13	2	0	17
RC106-100	9	1507.7	1458.1	1458.10	0.00	6319.0	3707.9	1131	428	0	14
RC107-100	9	–	1315.3	1301.60	1.04	TL	1402.6	671	0	0	13
RC108-100	9	1322.8	1220.7	1190.60	2.47	TL	1882.4	3	0	0	12
RC201-100	2	1836.0	1598.1	1598.10	0.00	4537.7	4395.0	131	40	1	13
RC202-100	2	1417.1	1364.8	1345.90	1.38	TL	584.3	142	0	0	13
RC203-100	2	1276.2	1172.4	1172.40	0.00	831.7	831.6	1	0	0	10
RC204-100	2	1013.5	951.0	951.00	0.00	4809.1	4809.0	1	0	0	8
RC205-100	2	1516.6	1387.2	1387.20	0.00	2978.9	2898.4	23	0	0	12
RC206-100	2	1411.2	1320.6	1306.10	1.10	TL	1931.0	166	0	0	11
RC207-100	2	1250.4	1157.8	1157.80	0.00	890.9	799.7	1	1	0	10
RC208-100	2	973.6	950.5	942.30	0.86	TL	6591.1	2	0	0	8